way as the objects in our everyday lives do. Therefore, they claim, it is impossible to talk about reality in quantum physics. There is not, nor could there be, any story of the world that goes along with the theory.

The popularity of this attitude to quantum physics is surprising. Physics is about the world around us. It aims to understand the fundamental constituents of the universe and how they behave. Many physicists are driven to enter the field out of a desire to understand the most basic properties of nature, to see how the puzzle fits together. Yet, when it comes to quantum physics, the majority of physicists are perfectly willing to abandon this quest and instead merely "shut up and calculate," in the words of physicist David Mermin.

More surprising still is that this majority view has, time and again, been shown not to work. Despite the popular view among physicists, Einstein clearly got the better of Bohr in their debates and convincingly showed there were deep problems that needed answering at the heart of quantum physics. Simply dismissing questions about reality as "unscientific," as some of Schrödinger's opponents did, is an untenable position based on outdated philosophy. And some dissenters from the majority have developed alternative approaches to quantum physics that clearly explain what is going on in the world without sacrificing any of the theory's accuracy.

The existence of these viable alternatives puts the lie to the idea that we are forced to give up on reality in quantum physics. Yet most physicists still subscribe to some form of this idea. It's still what's taught in classrooms, and it's still the picture that's usually painted for the public. Even when the alternatives are mentioned, they are mentioned as just that—alternatives to the default, despite the fact that the default is entirely unworkable. Thus, nearly a century after quantum theory was first developed—after it has thoroughly altered the world and the lives of every single human in it, both for better and worse—we still don't know what it's telling us about the nature of reality. This thoroughly strange story is the subject of this book.

This is an astonishing state of affairs, and hardly anyone outside of physics